

Homework 2 Question 4

a) No strategy is strictly dominated.

Proof 1: A strategy s_i is strictly dominated if there exists another strategy $s'_i \in S_i$ such that for all $s_{-i} \in S_{-i}$, $u_i(s'_i, s_{-i}) > u_i(s_i, s_{-i})$. Thus, to show that s_i is not strictly dominated, we have that find a $s_{-i} \in S_{-i}$ such that $u_i(s_i, s_{-i}) \geq u_i(s'_i, s_{-i})$ for all $s'_i \in S_i$. Three cases have to be considered:

Case 1: $s_i < v_i$ is not strictly dominated. Let $s_{-i} = s_i - 1$. Then we have $u_i(s_i, s_{-i}) = v_i - s_i$, $u_i(s_i - 1, s_{-i}) = \frac{v_i - s_i + 1}{2}$, $u_i(s_i - k, s_{-i}) = 0$ for all $k \in \mathbb{N}$ and $k \geq 2$, and $u_i(s_i + l, s_{-i}) = v_i - s_i - l$ for all $l \in \mathbb{N}$ and $l \geq 1$. Thus we have $u_i(s_i, s_{-i}) \geq u_i(s'_i, s_{-i})$ for all $s'_i \in S_i$.

Case 2: $s_i = v_i$ is not strictly dominated. Let $s_{-i} = v_i$. Then $u_i(v_i, s_{-i}) = 0$, $u_i(s'_i, s_{-i}) < 0$ if $s'_i > v_i$, $u_i(s'_i, s_{-i}) = 0$ if $s'_i < v_i$. Again, we have $u_i(s_i, s_{-i}) \geq u_i(s'_i, s_{-i})$ for all $s'_i \in S_i$.

Case 3: $s_i > v_i$ is not strictly dominated. Let $s_{-i} = s_i + 1$. Then $u_i(s_i, s_{-i}) = 0$, $u_i(s'_i, s_{-i}) < 0$ if $s'_i > s_{-i}$, $u_i(s'_i, s_{-i}) = 0$ if $s'_i < v_i$.

Proof 2: if some $s_i \in S_i$ is strictly dominated, then there exists some $s'_i \neq s_i$ such that for all $s_{-i} \in S_{-i}$, we have

$$u_i(s'_i, s_{-i}) > u_i(s_i, s_{-i}). \quad (1)$$

But (1) cannot hold because if $s_{-i} = \max\{s'_i, s_i\} + 1$, then

$$u_i(s'_i, s_{-i}) = 0 = u_i(s_i, s_{-i}).$$

b) Yes, weakly dominated strategies exist in the following 4 cases:

A strategy s_i is weakly dominated if there exists another strategy $s'_i \in S_i$ such that for all $s_{-i} \in S_{-i}$, $u_i(s'_i, s_{-i}) \geq u_i(s_i, s_{-i})$ and for some $s_{-i} \in S_{-i}$, $u_i(s'_i, s_{-i}) > u_i(s_i, s_{-i})$.

Case 1: $s_i > v_i$ is weakly dominated by v_i for player i . The reason goes as follows: $u_i(v_i, s_{-i}) = 0$, $u_i(s_i, s_{-i}) = 0$ when $s_{-i} > s_i$, $u_i(s_i, s_{-i}) < 0$ when $s_{-i} \leq s_i$.

Case 2: And $s'_i = 0$ is weakly dominated by $s_i = 1$ when $v_i > 2$. This can easily be found from payoffs:

$$u_i(s_i, s_{-i}) = \begin{cases} v_i - 1, & \text{if } s_{-i} = 0; \\ \frac{v_i - 1}{2}, & \text{if } s_{-i} = 1; \\ 0, & \text{if } s_{-i} > 1. \end{cases} \quad u_i(s'_i, s_{-i}) = \begin{cases} \frac{v_i}{2}, & \text{if } s_{-i} = 0; \\ 0, & \text{if } s_{-i} = 1; \\ 0, & \text{if } s_{-i} > 1. \end{cases}$$

Case 3: if $v_i = 1$, then $s'_i = 1$ is weakly dominated by $s_i = 0$ because

$$u_i(s_i, s_{-i}) = \begin{cases} \frac{1}{2}, & \text{if } s_{-i} = 0; \\ 0, & \text{if } s_{-i} = 1; \\ 0, & \text{if } s_{-i} > 1. \end{cases} \quad u_i(s'_i, s_{-i}) = \begin{cases} 0, & \text{if } s_{-i} = 0; \\ 0, & \text{if } s_{-i} = 1; \\ 0, & \text{if } s_{-i} > 1. \end{cases}$$

Case 4: if $v_i = 2$, then $s'_i = 2$ is weakly dominated by $s_i = 1$ because

$$u_i(s_i, s_{-i}) = \begin{cases} 1, & \text{if } s_{-i} = 0; \\ \frac{1}{2}, & \text{if } s_{-i} = 1; \\ 0, & \text{if } s_{-i} > 1. \end{cases} \quad u_i(s'_i, s_{-i}) = \begin{cases} 0, & \text{if } s_{-i} = 0; \\ 0, & \text{if } s_{-i} = 1; \\ 0, & \text{if } s_{-i} > 1. \end{cases}$$

c) Nash equilibria are not unique, and can be find by enumerating and comparing $u_i(\cdot)$.

Case 0: $v_i = v_{-i} = v$. NE are $(s_i, s_{-i}) \in \{(v, v), (v-1, v-1), (v-2, v-2)\}$.

Case 1: $v_i = v_{-i} + 1 = v + 1$. NE are $(s_i, s_{-i}) \in \{(v, v), (v-1, v-1), (v, v-1)\}$.

Case 2: $v_i = v_{-i} + 2 = v + 2$. NE are $(s_i, s_{-i}) \in \{(v, v), (v, v-1), (v+1, v)\}$.

Case 3: $v_i = v_{-i} + 3 = v + 3$. NE are $(s_i, s_{-i}) \in \{(v, v-1), (v+1, v), (v+2, v+1)\}$.

Case 4: $v_i = v_{-i} + k = v + k$, where $k \in \mathbb{N}$ and $k \geq 4$. NE are given by

$$(s_i, s_{-i}) \in \{(v, v-1), (v+1, v), (v+2, v+1), \dots, (v+k-1, v+k-2)\}$$

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